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MILITARY SPECIFICATION

TRANSMISSION, CROSS-DRIVE: COMPLETE ASSEMBLY; CENTER SECTION; GEAR ASSEMBLIES, OUTPUT REDUCTION

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the XT-1410-4 transmission composed of a center section, and a right hand and left hand final drive reduction assembly, for use in military vehicles (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2520

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
MILITARY

MIL-L-2104	- Lubricating Oil, Internal-Combustion Engine, Tactical Service.
MIL-L-21260	- Lubricating Oil, Internal-Combustion Engine, Preservative, and Break-in.
MIL-L-46167	- Lubricating Oil, Internal-Combustion Engine, Arctic.

STANDARDS
MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-45662	- Calibration System Requirements.

2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

8350912	- Reduction Assembly, Left Hand Output, Transmission XT-1410-2.
8350913	- Reduction Assembly, Right Hand Output, Transmission XT-1410-2.
11649957	- Procedure for Testing of XT-1410 Series Output Reduction Assemblies.
11649999	- Transmission Assembly, Installation.
11650001	- Center Section Assembly - Transmission XT-1410-4.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish transmission assemblies which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent assemblies delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Reliability. During Government specified vehicle tests, the transmission assembly shall be installed in the vehicle for which it is being procured. It shall require no repair and no maintenance, other than as specified in 3.2.1, during vehicle operation at a Government proving ground for distances up to 4,000 miles. Operation shall include any of the types of operation, terrain conditions and ambient temperatures to which a military vehicle of the kind is subjected in accordance with the provisions of the specification covering the vehicle (see 4.6.5).

3.2.1 Maintenance. During the operation specified in 3.2, the permissable maintenance shall consist of filter changes, oil additions, oil changes and brake adjustments.

3.3 Materials. Materials shall be as specified herein and in referenced specifications, standards and drawings. External surfaces shall be corrosion resistant or shall be properly protected to prevent corrosion. Materials shall be free from defects which adversely affect performance or serviceability of the finished product (see 4.1.2 and 6.4).

3.4 Design and construction. The transmission assembly design and construction shall be in accordance with Drawings 8350912, 8350913, 11649999 and 11650001. All parts requiring identification shall be marked in accordance with MIL-STD-130 (see 4.1.2).

3.4.1 Oil. Oil conforming to type PE10-1 of MIL-L-21260 shall be used during transmission tests and production runs. Oil conforming to MIL-L-46167 shall be used when the transmission is operated in ambient temperatures below zero degrees Fahrenheit (°F). Oil conforming to OE/HDO-10 of MIL-L-2104 shall be used after installation of the transmission in its intended vehicle, when operating in ambient temperatures above minus 10°F (see 4.1.2).

3.5 Performance.

3.5.1 Functional operation. All shift and steer positions of the transmission shall function without any indication of mechanical malfunction. The oil level in the transmission shall be maintained within the operating range on the oil level indicator (dip stick) (see 4.6.1).

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3.5.2 Center section operation.

3.5.2.1 Output shaft stalled. With output shafts stalled and a torque input of 740 ± 5 pounds-feet (lb-ft) the speeds, torques, oil pressures and oil flows as applicable shall be within the limits specified in table I (see 4.6.3.1 through 4.6.3.4).

3.5.2.2 Third range no load. With the transmission in 3rd range, an input speed of 2350 ± 10 rpm and no load on the output shaft, the input torque and oil pressures shall conform to applicable values shown in table I (see 4.6.3.5).

3.5.2.3 Neutral range steer oil pressure. With the range selector in neutral, the transmission oil pressures, measured at the steer clutch (RH or LH) and lube positions, when full left steer or full right steer is applied, shall be not less than the oil pressure indicated as follows (see 4.6.3.6).

a.	Steer clutch (RH or LH)	105 psi
b.	Lube	3 psi

3.5.2.4 Lockup clutch. With the transmission in third range the transmission output speeds at the time of lockup clutch engagement and disengagement shall be in accordance with the following (see 4.6.3.7):

a. Lockup engages at 1290 to 1430 output speed (rpm).
b. Lockup releases at 1080 to 1225 output speed (rpm).

NOTE: The output speed is an average of the RH and LH output speeds.

3.5.2.5 Brake coolant pump. With the brake coolant pump air valve closed, the transmission in third range, 1500 ± 25 rpm input speed and no load on the output shafts, the oil pressure in both left and right output support fixtures (see 4.6.2) shall show an indication of pressure above atmospheric (see 4.6.3.8).

3.5.2.6 Neutral range torque. With the transmission in neutral range, the output shafts stalled, a sump temperature of 200 ± 25°F and an input speed of 2200 ± 10 rpm, the input torque shall be no more than 120 lb-ft (see 4.6.3.9).

3.5.2.7 Down shift inhibitor speed range. The transmission input speed at the point where shifts in gear range is possible, shall be within the following speed ranges as applicable. Downshifts at speeds above the range shall not be possible (see 4.6.3.10).

	Range shift	Input speed (rpm)
a.	Third to second	1150 to 1400
b.	First to reverse	1000 to 1400

3.5.2.8 Leakage. The transmission shall show no evidence of a leak. A weep or a seep is permissible (see 4.6.3.11). See definitions of leakage in paragraph 6.3.2.

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TABLE I. Load and no load values.

Gear range selector position	Requirement	Test	Input		Output		Oil pressure psi			Lockup clutch apply	Cooler oil flow gpm	Converter oil temp °F
			Speed rpm	Torque lb-ft	Speed rpm	Torque lb-ft	Main	Converter in	Lub oil			
3rd stall	3.5.2.1	4.6.3.1	1260-1370	*740 ± 5	Stall	3300 min	118 min	10 min	12 min		26 min	*240-260
2nd stall	3.5.2.1	4.6.3.2	1260-1370	*740 ± 5	Stall		118 min					*240-260
1st stall	3.5.2.1	4.6.3.3	1240-1370	*740 ± 5	Stall		275 min					*240-260
Reverse	3.5.2.1	4.6.3.4	1240-1370	*740 ± 5	Stall		275 min					*240-260
3rd no-load	3.5.2.2	4.6.3.5	*2340-2360	160 max			210 max	140 max	18 min	**		*240-260

NOTE:

* Pre-determined input values.
** Minimum clutch apply pressure to be not less than 5 psi below the actual main oil pressure.

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3.5.3 Final drives operation.

3.5.3.1 Operation. The input torque at 1800 ± 10 rpm input speed with no load on the output shaft shall not be more than 37 lb-ft (see 4.6.4.1).

3.5.3.2 Leakage. The final drives shall show no evidence of an external oil "leak" or "drip". A "weep" or a "seep" is permissible (see 4.6.4.2). See 6.3.2 for the leakage definitions.

3.5.3.3 Service brake. The service brake with 80 lb-ft torque applied to the brake apply shaft, shall withstand a 14,000 lb-ft torque applied to the final drive output shaft, without slipping of the output shaft (see 4.6.4.3).

3.6 Exterior surface. All exposed surfaces of the transmission and its components shall be cleaned, painted, or treated for corrosion resistance as specified on the applicable drawings, or if not so specified, in accordance with the applicable provisions of MIL-STD-193.

3.7 Identification and marking. The complete transmission and each major component (center section, final drive reduction assemblies) shall be marked in accordance with MIL-STD-130 as specified on applicable drawings.

3.8 Workmanship. The workmanship shall be in accordance with manufacturing practice for the product covered by this specification and shall be such as to produce tranmissions free of defects that would affect their serviceability and shall conform to requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract, the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Selected commercial, modified commercial, contractor designed inspection equipment or measuring set-ups must be capable of repetitive measurements within an accuracy of 10 percent of the measured dimension allowable tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

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4.1.2 Materials and construction. To determine conformance to 3.3 through 3.4.1, inspection and material certification records shall be maintained by the contractor. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Acceptance tests (see 4.5.3.1).
   3. Control tests (see 4.5.3.2).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions:

a. Air temperature 75° ± 10°F (23° ± 10°C)
b. Barometric pressure 28.54^{+2}_{-3} inches mercury (Hg)
c. Relative humidity 50 ± 30 percent

4.3.1 Calibration of test equipment. Unless otherwise specified, test equipment accuracy shall be capable of measuring within 10 percent of product test tolerance. Calibration of test equipment shall be conducted at intervals which serve to establish required accuracy. Records of calibration shall be made available to the Government. The Government inspector shall refuse to allow inspection where accuracy of test equipment has not been established to the satisfaction of the Government in accordance to requirements of MIL-STD-45662.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select three transmission assemblies produced under the production contract (see 6.2) for first article inspection. First article samples shall be examined as specified in 4.5.2 and tested as specified in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply transmission assemblies that are fully representative of those inspected and accepted as a first article sample. Any changes or deviation of the production units from the accepted first article sample shall be subject to the approval of the contracting officer. One accepted first article sample will remain at the contractor to serve as a standard for subsequent production, it will be the last one delivered on the contract.

4.4.1 First article inspection failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval until corrective action by the contractor has been approved by the Government.

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TABLE II. Classification of inspections.

Title	Requirement	Test	First article	Quality conformance	
				Acceptance (100%)	Control
Transmission center section:					
Functional operation	3.5.1	4.6.2	X	X	X
Output shaft third range stalled	3.5.2.1	4.6.3.1	X	X	X
Output shaft second range stalled	3.5.2.1	4.6.3.2	X	X	X
Output shaft first range stalled	3.5.2.1	4.6.3.3	X	X	X
Output shaft reverse stalled	3.5.2.1	4.6.3.4	X	X	X
Third range no load	3.5.2.2	4.6.3.5	X	X	X
Neutral range steer oil pressure	3.5.2.3	4.6.3.6	X	X	X
Lockup clutch	3.5.2.4	4.6.3.7	X	X	X
Brake coolant pump	3.5.2.5	4.6.3.8	X	X	X
Neutral range torque	3.5.2.6	4.6.3.9	X	X	X
Down shift inhibitor speed range	3.5.2.7	4.6.3.10	X	X	X
Leakage	3.5.2.8	4.6.3.11	X	X	X
Final drives:					
Operation	3.5.3.1	4.6.4.1	X	X	X
Leakage	3.5.3.2	4.6.4.2	X	X	X
Service brake	3.5.3.3	4.6.4.3	X	X	X

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the transmission assemblies from an identifiable production period, from one manufacturer, produced under essentially the same conditions, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Quality conformance examinations.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL) on the basis of percent defective.

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Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

4.5.3 Quality conformance tests.

4.5.3.1 Acceptance tests (100 percent). Each transmission assembly shall be subjected to the tests specified in table II.

4.5.3.2 Control tests. Control tests shall be conducted on one transmission assembly from each lot of 200 units consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. The transmission assemblies shall be subjected to the tests specified in table II.

4.5.4 Failure. Failure of any transmission assembly to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Dimensions affecting interchangeability, out of tolerance.	SIE 1/, and Visual
102	Control levers and valves, malfunction	Functional and Visual
103	Hydraulic lines and fittings, improper installation.	Functional and Visual
104	Oil filters, dirt or foreign matter plugging lines and screens.	Functional and Visual
Minor	AQL 2.5% Defective	
201	Dimensions not affecting interchangeability.	SIE 1/, and Visual
202	Identification and marking, improper or not legible.	SIE 1/, and Visual
203	Appearance, scratches, chipped, pitted or burred.	Visual
204	Paint-finish coat, improper application or color.	Visual
205	Workmanship, faulty, not following good practice, improper installation or adjustment of components.	Functional and Visual

1/ SIE = Standard Inspection Equipment.

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4.6 Methods of inspection. Unless otherwise specified all transmission assembly tests specified herein shall be conducted with the applicable oil for the ambient temperature of the specified test condition (see 3.4.1).

4.6.1 Test conditions. To determine conformance to requirements of 3.5 through 3.5.3.3, the transmission assembly center section shall be installed on an absorption dynamometer test stand 1/. NOTE: Provisions shall be made for connecting the input and output shafts, range shift controls, oil pressure and temperature outlets. The transmission shall be filled with oil so that the oil level is between the "ADD" and FULL" marks on this oil level indicator (dip stick). Instrumentation shall be provided to measure and record the following:

a. Input speed - rpm.
b. L.H. output speed - rpm.
c. R.H. output speed - rpm.
d. Input torque - lb.-ft.
e. L.H. output torque (at stall) - lb.-ft.
f. R.H. output torque (at stall) - lb.-ft.
g. Main oil pressure - psi.
h. Converter in oil pressure - psi.
i. Lube oil pressure - psi.
j. Lockup clutch apply pressure - psi.
k. Governor oil pressure - psi.
l. Simulated engine manifold pressure - 0.20 in. of mercury (variable).
m. 1st gear apply pressure - psi.
n. 2nd gear apply pressure - psi.
o. 3rd gear apply pressure - psi.
p. Reverse gear apply pressure - psi.
q. L.H. steer clutch pressure - psi.
r. R.H. steer clutch pressure - psi.
s. Converter out oil temperature - (°F).
t. Sump oil temperature - (°F).
u. Cooler oil flow - gallons per minute (gpm).
v. Sump oil level - sight gage indicating oil level in reservoir (optional).
w. Brake coolant oil pressure - psi.

4.6.2 Functional test. To determine conformance to 3.5.1, with the transmission in third range and output shafts stalled, the transmission shall be operated at 1,000 ± 20 rpm input speed for 2 minutes. The output shafts shall be released and the transmission allowed to free run at 300 to 400 rpm for 3 minutes. While operating at 1000 rpm input speed, all shaft and steer positions shall be checked for indication of noise, drag, mechanical malfunction, and low oil pressure. Stop the transmission, wait 3 to 5 minutes and recheck oil level and adjust if necessary. It is optional to fill transmission with a measured quantity of oil to produce the same operating level as determined in 4.6.1.

1/ The right and left hand support fixtures of the test stand must provide sealed closures for the transmission trunnions. The closures must permit access for installation of stall devices and provide provisions for measuring brake coolant oil pressure.

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4.6.3 Center section tests.

4.6.3.1 Third range stall test. The transmission output shafts shall be coupled to the absorption dynamometers and the transmission placed in third range. The transmission shall then be operated with the output shafts stalled by the test stand stall devices, to determine conformance to 3.5.2.1. Applicable data and measurements as required shall be recorded.

4.6.3.2 Second range stall test. The transmission output shafts shall be coupled to the absorption dynamometers and the transmission placed in second range. The transmission shall then be operated, with the output shafts stalled to determine conformance to 3.5.2.1. Applicable data and measurements as required shall be recorded.

4.6.3.3 First range stall test. The transmission output shafts shall be coupled to the absorption dynamometers and the transmission placed in first range. The transmission shall then be operated, with the output shafts stalled to determine conformance to 3.5.2.1. Applicable data and measurements as required shall be recorded.

4.6.3.4 Reverse range stall test. The transmission output shafts shall be coupled to the absorption dynamometers and the transmission placed in reverse range. The transmission shall then be operated, with the output shafts stalled to determine conformance to 3.5.2.1. Applicable data and measurements as required shall be recorded.

4.6.3.5 Third range no load test. The transmission output shafts shall be coupled to the absorption dynamometers and the transmission placed in third range. The transmission shall then be operated to determine conformance to 3.5.2.2. Applicable data and measurements as required shall be recorded.

4.6.3.6 Neutral range steer oil pressure. The transmission shall be placed in neutral range and shifted gradually into full right steer while at 800 rpm input speed. The transmission shall then be operated with left hand output shaft in forward rotation and right hand output shaft in reverse rotation to determine conformance to 3.5.2.3. Applicable data and measurements as required shall be recorded.

4.6.3.7 Lockup clutch test. To determine conformance to 3.5.2.4, the transmission ouput shafts shall be coupled to the absorption dynamometers and the transmission placed in third range, the output load released and converter out oil temperature stabilized at 250 $\pm$ 10°F. Increase input speed until the lockup clutch engages. Reduce transmission output speed until lockup clutch disengages applicable data and measurements as required shall be recorded.

4.6.3.8 Brake coolant pump test. The brake coolant pump air valve shall be closed with the transmission in third range, and no load on the absorption dynamometers. The transmission shall then be operated with an input speed of 1500 $\pm$ 25 rpm and the oil pressure in both left and right output support fixtures recorded to determine conformance to 3.5.2.5.

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4.6.3.9 Neutral range torque test. The transmission shall be placed in neutral position and output shafts shall be coupled to the absorption dynamometers and stalled. With all clutch pressures at 0 psi, the transmission shall be operated and the input torque recorded to determine conformance to 3.5.2.6.

4.6.3.10 Down shift inhibitor test. To determine conformance to 3.5.2.7, the output shafts shall be disconnected from the absorption dynamometers. With the transmission in third range, the converter oil out temperature at 180 ± 20°F, and with an input speed of 2000 rpm, gradually reduce the input speed to the point where a shift to second range can be made. With the transmission in first range and an input speed of 1500 rpm, gradually reduce the input speed to the point where a shift into reverse range can be made. The input speed at which each shift can be made shall be recorded.

4.6.3.11 Leakage test. To determine conformance to 3.5.2.8, the exterior surfaces of the transmission shall be cleaned and dried. Oil filters shall be removed, cleaned and reinstalled, pressure gage fittings shall be removed and replace with pipe plugs specified on Drawing 11649999. With no load on the output shafts, 1000 ± 20 rpm input speed and 200 to 280°F converter out oil temperature, the transmission shall be operated in all range and steer positions for not less than 15 minutes. Inspection shall be made for external leaks.

4.6.4 Final drive test conditions. To determine conformance to requirements of 3.5.3, the final drive assembly shall be coupled to the test stand. An auxiliary oil pump, and a monitored and controlled oil supply system shall be provided as part of the test stand equipment. The oil pump shall have a flow capacity in the range of 2 to 4 gpm at a temperature of plus 180 to 200°F which shall be introduced through the service brakes before and during running of the final drives. A sufficient amount of air shall be introduced with the oil to provide scavenging of oil from the unit. All necessary connections shall be made to the input dynamometer, output absorption dynamometer, service brakes, lubrication oil, and brake cooling oil. Service brakes shall be adjusted in accordance with method specified on Drawing 11649957. Instrumentation shall be provided to measure speeds and torques:

a. Input speed - rpm.
b. Input torque - lb-ft

4.6.4.1 Final drive operation test. To determine conformance to 3.5.3.1, the final drive output shafts shall be connected to the absorption dynamometer and the final drive output assembly shall be operated at an input speed of 1800 rpm for 5 minutes with an absorption dynamometer load on the output of the final drive so as to have 1000 lb.-ft. of input torque. The final drive output shafts shall then be disconnected from the output absorption dynamometer, the final drive assembly shall be operated, and the input torque shall be recorded.

4.6.4.2 Final drive leakage test. To determine conformance to 3.5.3.2, a check of the exterior of the final drives shall be made for leakage while conducting the tests specified in 4.6.4.1 and 4.6.4.3.

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4.6.4.3 Service brake test. The input and output dynamometers shall be disconnected and the torque arm assembly installed to the final drive outputs. With 80 lb.-ft. torque applied to the service brake apply shaft, the torque shall be increased on the output shaft of the final drive by means of the torque arm assembly until the service brakes slip or until 22000 lb.-ft. torque has been applied without slippage. Record the output torque applied to determine conformance to 3.5.3.3.

4.6.5 Reliability test. To determine conformance to 3.2, the installed transmissions shall be observed during the vehicle test.

4.7 Exterior surface. To determine conformance to 3.6 all transmission assemblies shall be examined to verify condition of exterior surfaces as specified on the applicable drawings or applicable provisions of MIL-STD-193.

4.8 Identification and marking. The complete transmission assembly shall be visually examined to verify conformance to the identification and marking requirements of 3.7.

4.9 Workmanship. Workmanship examinations shall be performed phases of the assembly process to verify conformance with 3.8.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desire level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The complete transmission and major component assemblies covered by this specification are intended for use as original equipment, as spares, or as replacements in military recovery vehicle, full tracked, M88A1.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Whether complete transmission or specific component (center section, right, or left hand final drive) is required (see 1.1).
c. Type required (see 1.2).
d. Whether preproduction samples shall be furnished (see 4.4).
e. Whether initial production units shall not be furnished (see 4.4).
f. Selection of applicable level and packaging standard or packaging data sheet (see 5.1).

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6.3 Definitions.

6.3.1 Transmission orientation. The following definitions are established relative to the transmission and are not to be confused with vehicle definitions:

a. "Front" of the transmission mounts next to the engine.
b. "Rear" is the opposite end.
c. "R.H." and "L.H." is to the viewer's right and left, respectively, as he stands at the rear facing the transmission assembly.

6.3.2 Leakage. The following definitions are established as guide in determining conformance with the leakage requirements fo this specification.

a. "Weep" - Any evidence of fluid beyond a seal or splitline.
b. "Seep" - Any evidence of fluid beyond a seal or splitline that does not result in the formation of a droplet.
c. "Leak" - Any evidence of fluid beyond a seal or splitline that results in the formation of a droplet.
d. "Drip" - Any evidence of fluid beyond a seal or splitline where droplets form and fall.

6.4 Recycled materials. Use of recycled materials which meet the requirments of applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.3).

Custodian:
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